MILITARY SPECIFICATION

VALVE, CHARGING

This specification is approved for use by Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the design, manufacture, inspection and testing of charging valves.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-N-411	Nitrogen, Technical.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD} Code 93, Lakehurst, New Jersey 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4820

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MILITARY

DOD-D-1000	Drawings, Engineering and Associated Lists.
MIL-B-22020	Bag, Transparent, Flexible, Sealable, Volatile Corrosion Inhibitor, Treated.

STANDARDS

MILITARY

MIL-STD-109	Quality Assurance Terms and Definitions.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of.
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of.
MIL-STD-810	Environmental Test Methods.
MIL-STD-831	Test Reports, Preparation of.
MS 20813	Cap, High Pressure Air Valve.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND
(Code Ident 30003)

DL 58A164D556	Receiver Assembly, Nitrogen.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.4 and 6.4).

3.1.1 First article sample. Unless otherwise specified in the contract or purchase order (see 6.2.1), a first article sample of the charging valve shall be delivered to the testing activity designated in the contract or purchase order (see 6.2.1). The first article sample shall be manufactured using the same methods, materials, processes, and procedures proposed for production. Any production prior to acceptance of the first article sample is at the contractor's risk.

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3.2 Selection and quality verification of parts and materials. Quality verified parts (see 3.2.1] and materials from approved sources (see 3.2.2) shall be used in the fabrication of the end item. The contractor shall prepare reports of conformance to applicable requirements.

3.2.1 Approval of parts and materials. Parts and materials proposed for use in the fabrication of the end item that are not specified in detail herein or in applicable referenced documents shall be selected from other documents in the order of precedence specified in MIL-STD-143 and shall be approved by the procuring activity prior to use. Approval shall be predicated upon source (vendor) approval and verification of part quality by test.

3.2.2 Source approval. The contractor shall be responsible for ascertaining that each proposed source for parts and materials maintains a quality control program to ensure consistent delivery of products conforming to applicable requirements. When parts or materials are fabricated by the contractor, the area used for that purpose shall be considered the same as *an* outside vendor insofar as source approval is concerned. Source approvals by the contractor shall be subject to concurrence by the procuring activity. Source approval will be granted by the procuring activity upon preparation of acceptable evidence of any of the following:

a. Formal on-site survey of the vendor's manufacturing facilities.

b. The source has been previously approved, and evidence is available to confirm that acceptable product quality has been maintained.

c. Other evidence of acceptable product quality is available.

3.2.3 verification of part quality. Quality verification shall be demonstrated by successful testing of the part to levels of environmental, electrical, and physical stresses equal to or greater than those which will be encountered in the system and as required by the part specification. Part quality may be demonstrated by any of the following:

a. When the part is manufactured and qualified to a military specification which includes requirements equal to or more severe than those of the end item application.

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b. When the part is manufactured and qualified to a military specification not including all requirements of the application, and additional tests for such requirements have been performed.

c. When the part is not qualified to a military specification, but all tests appropriate to the application have been performed.

3.2.4 Approved source/quality verified parts list (AS/QVPL). An AS/QVPL shall be prepared by the contractor for all subassemblies and parts for approval by the procuring activity. All entries on the AS/QVPL shall be fully substantiated by the preparation of data to satisfy the requirements of source approval and part test approval as specified herein. Reports of all quality verification testing shall conform to the requirements of MIL-STD-831. The contractor may solicit from the procuring activity information as to the status of quality verified parts and approved sources which may have been previously developed. Neither part environmental tests nor part environmental test reports are required in those instances where vendors are selected from current AS/QVPLs of other programs. Entries on the AS/QVPL shall identify the vendor and the product which is proposed for use in the charging valves by the contractor. In the event several sources have previously been approved for the application, the contractor shall specify the source of the parts to be used. Each item listed shall be supported in accordance with the previsions of this section. This listing does not supplant or modify the applicable documentation or other Government Qualified Parts Listing.

3.2.4.1 Conditions governing the quality verification of parts. When the contractor proposes to demonstrate the quality verification of parts, as provided herein, the following provisions shall apply:

a. Test may be performed by the contractor, utilizing his own facilities, the facility of the manufacturer of the part or by any other suitable facility selected by the contractor. In any event, the facilities shall be approved by the contractor prior to the performance of any test. This information shall be prepared for approval by the procuring activity.

b. Prior to the performance of any test, a detailed test procedure shall be prepared by the contractor and approved by the procuring activity. The procedure shall identify, as a minimum, the step by step functions to be performed and the test equipment to be used, and shall provide for full conformance with the provisions of the applicable documentation. The Government reserves the right to review any of the individual test procedures at its discretion.

c. Quality verification testing may be monitored by a representative of the procuring activity.

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3.2.4.2 Changes to the AS/QVPL. The contractor may develop alternative sources for any part, subject to the provisions herein. Proposals by the contractor for part substitution or changes to the AS/QVPL shall include proof that the proposed substitute part will meet the requirements of 3.2.2 and 3.2.3. The AS/QVPL shall be revised and updated as applicable to include approved changes.

3.2.4.3 Removal from the AS/QVPL. Evidence of poor quality may be cause for suspension or removal of a source from the AS/QVPL at the OPtiOn Of the contractor or the procuring activity. The contractor shall develop quality verified sources for those parts for which the approved source has been removed. Quality reverification shall be required under any of the following conditions unless otherwise directed by the procuring activity:

a. When the part has been modified by a change in design, materials, manufacturing process, manufacturing inspection procedures, or manufacturing facilities. The procuring activity shall be notified prior to delivery of any parts incorporating any such changes and be provided quantitative evidence of the effect of such changes on the performance or other characteristics of the charging valve.

b. When there is evidence that the quality of the part has not been maintained. This evidence may be in the form of accumulated failure reports which indicated that the quality of the part has deteriorated, or there is a failure trend.

c. When the manufacturer has not produced the item for 6 or more months.

3.2.5 Part conformance provisions. When a drawing or specification for a part invokes by reference the provisions of a military specification, total conformance to the provisions of the referenced specification shall be required. Total conformance is defined as conformance to the stated performance and product characteristics, and compliance with all quality assurance provisions, including qualification, quality verification, and quality conformance testing.

3.3 Design and construction. The design and construction of the charging valves shall be in accordance with the applicable drawings listed on DL 58A164D556 and as specified herein. The contractor shall notify the procuring activity of any changes in design and construction after the charging valve has been accepted by the procuring activity.

3.3.1 Materials. Materials shall be selected on the basis of strength, workability and compatibility with other materials in the item so as to minimize galvanic action, corrosion, etc.

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3.3.2 Finishes. Finishes shall be applied to all materials, whenever applicable, and shall be for the purpose intended, such as corrosion resistance, increased durability, etc. External surfaces shall be in accordance with the applicable drawings listed on DL 58A164D556.

3.3.3 Interchangeability. All charging valves having the same part number shall be internally and externally identical. Changes in part numbers shall be governed by drawing number requirements of DOD-D-1000.

3.3.4 Identification of parts. All charging valves shall be marked as specified on the applicable drawings listed on DL 58A164D556 or in accordance with MIL-STD-130 if not specified on the drawing.

3.3.5 Dimensions. The charging valve envelope dimensions shall be in accordance with the applicable drawing for the charging valve being procured.

3.4 Performance. The charging valve, when constructed in accordance with the applicable drawings list on DL 58A164D556, shall meet the following requirements:

3.4.1 Cleanliness. The charging valve shall be free of all water, oils, grease and foreign matter exceeding 0.002 inches in size on all surfaces.

3.4.2 Proof pressure. The charging valve shall operate satisfactorily after being subjected to 5500 ±100 psig for three (3) minutes. There shall be no permanent deformation.

3.4.3 Leakage. The charging valve shall not have any leakage in excess of 0.2 cubic centimeters per minute after being pressurized to each of the following pressures for three (3) minutes:

a. 200 psig

b. 1000 psig

c. 2200 psig

d. 3000 psig

e. 3800 psig

3.4.4 Operating pressure. The charging valve shall operate satisfactorily when the inlet port is pressurized to 3800 ±100 psig and the inlet pressure line is removed. Leakage shall not exceed 0.2 cubic centimeters per minute and the 3800 psig shall remain in the test fixture.

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3.4.5 Pressure drop. The pressure drop across the charging valve shall not exceed 10 psig with a maintained inlet pressure of 3000 ±100 psig while flowing at 28.3 liters per minute (one cubic foot per minute).

3.4.6 Environmental. The charging valve shall meet the following environmental requirements of MIL-STD-810, unless otherwise specified herein.

3.4.6.1 High temperature. The charging valve shall operate satisfactorily after the test of MIL-STD-810, Method 501, Procedure I, except the temperature shall be 121°C (250°F).

3.4.6.2 Low temperature. The charging valve shall operate satisfactorily after the test of MIL-STD-810, Method 502, Procedure I for a period of four hours.

3.4.6.3 Thermal shock. The charging valve shall operate satisfactorily after the test of MIL-STD-810, Method 503, Procedure I.

3.4.6.4 Dust. The charging valve shall operate satisfactorily after the test of MIL-STD-810, Method-510, Procedure I.

3.4.6.5 Salt fog. The charging valve shall operate satisfactorily after the test of M1L-STD-810, Method 509, Procedure I, except the duration shall be 50 hours.

3.4.6.6 Humidity. The charging valve shall operate satisfactorily after the test of MIL-STD-810, Method 507, Procedure I, except four 24 hour cycles shall be substituted for the ten 24 hour cycles.

3.4.6.7 Vibration. The charging valve shall operate satisfactorily after exposure to the sinusoidal and random vibration tests while pressurized to 3000 ±100 psig.

3.4.7 Service life. The charging valve shall operate satisfactorily after being subjected to a life test of 500 operating cycles and pressurized to 3000 ±100 psig as specified herein and there shall be no damage when visually inspected at the completion of each of the 50 cycles.

3.4.8 Burst pressure. The charging valve shall operate satisfactorily after being subjected to a burst pressure test wherein the outlet port shall be hydrostatically pressurized to 6700 ±100 psig for three (3) minutes with no evidence of permanent deformation.

3.5 Precedence. The selection of specifications and standards for use in the design and development shall be in accordance with the requirements of

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MIL-STD-143. When the requirements, this specification, and referenced documents are in conflict, the following precedence shall apply:

a. This specification. This specification shall have precedence over all referenced documents.

b. Referenced documents. Any documents referenced in this specification shall have precedence over all subsidiary documents referenced therein.

3.6 Workmanship. The workmanship displayed in fabrication and assembly of the charging valve shall be such as to assure, within design limitations, the ability of the charging valve to meet its performance requirements under all conditions specified herein. Particular attention shall be paid to neatness and thoroughness of assembly, marking of parts and assembly, plating, painting, machining, welding, brazing and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in the approved first article sample, subject to any qualification stated in the Government's notice of approval, shall determine the requirements of the remainder of the contract relative to workmanship.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure that the supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection of the charging valves shall be classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions, Unless otherwise specified herein, all inspections shall be performed under the following conditions and tolerances.

d. Room temperature. Room ambient of 25 ±10°C (77 ±18°F) indicated.

b. Test temperature. All parts of the test item shall be stabilized at the specified temperature ±2.8°C (±5°F) indicated prior to conducting any operational tests. Unless othewise specified

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herein, temperature stabilization will have been attained when the indicated temperature of the surface of the largest mass of the test item does not change more than ±2.8°C (±5°F) in a period of one hour.

c. Humidity. Room ambient to 95 percent relative humidity.

d. Altitude. Unless otherwise specified herein, testing shall be performed at normal ground elevation.

e. Test gas. The test gas shall be nitrogen in accordance with BB-N-411, Type 1, Class I, Grade A.

f. Test fixture. A high-strength, corrosion-resistant fixture shall be utilized for mating with the charging valve outlet port. The fixture shall have a charging valve port in accordance with DL 58A164D556 and an open port for gas transfer. The charging valve shall be torqued to the fixture. The test fixture shall be used for all tests of 4.6.3, 4.6.4, 4.6.5 and 4.6.6.

g. Torque requirements. The installation torque for the charging valve to the receiver assembly or test fixture shall be 275 ±25 inch-pounds. The closing torque shall be applied clockwise at 50 ±5 inch-pounds.

h. Volume and flow. When required, all volume and flow measurements shall be corrected to normal temperature and barometric pressure (NPT) conditions. NPT conditions are 760 millimeters (29.92 inches) of mercury and 21.1°C (70°F).

4.4 First article inspection. First article inspection shall consist of the tests outlined in Table I. The procuring activity reserves the right to delete tests and alter the test sequence of Table I (see 6.2.1). First article approval is valid only on the contract or purchase order under which it is granted, unless extended by the Government to other contracts or purchase orders.

4.4.1 Charging valve samples. First article charging valve samples shall consist of the first four charging valves manufactured on the contract or purchase order. The charging valves shall be permanently labeled for ease of identification and shall be produced with equipment and procedures that will be used for the production of all the charging valves designated in the contract or purchase order. Prior to submission, the contractor shall inspect the samples to the degree necessary to assure that they conform to the requirements of the contract and prepare a first article test report of this inspection with the samples, including statements of findings and certificates of conformance for materials, cleanliness, and tests A through F of Table I.

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TABLE I. First article inspections - schedule and sequence 1/.

Test	Test title	Test requirement paragraph	Test method paragraph	Sample A	Sample B	Sample C	Sample D
A	Cleanliness	3.4.1	4.6.1	1	1	1	1
B	Inspection	3.3.6	4.6.2	2	2	2	2
C	Proof pressure	3.4.2	4.6.3.1	3	3	3	3
D	Leakage	3.4.3	4.6.3.2	4	4	4	4
E	Operating pressure	3.4.4	4.6.3.3	5	5	5	5
F	Pressure drop	3.4.5	4.6.3.4	6	6	6	6
G	High temperature	3.4.6.1	4.6.4.1	7	x	7	x
H	Low temperature	3.4.6.2	4.6.4.2	8	x	8	X
I	Thermal shock	3.4.6.3	4.6.4.3	x	7	x	7
J	Dust	3.4.6.4	4.6.4.4	x	8	x	8
K	Salt fog	3.4.6.5	4.6.4.5	x	9	x	9
L	Humidity	“3.4.6.6	4.6.4.6	x	10	x	10
M	Vibration	3.4.6.7	4.6.4.7	9	x	9	x
N	Service life 2/	3.4.7	4.6.5	x	11	x	11
0	Burst pressure	3.4.8	4.6.6	10	x	10	x

1/ Test sequence is denoted by the numbers in the sample columns. An X denotes test not required.
2/ Charging valves subjected to life tests shall not be subjected to the burst test.

4.5 Quality conformance inspection. Acceptance of charging valves shall be on an inspection lot basis. Quality conformance inspection shall consist of the following tests:

a. Individual tests (see Table II, Group 1).

b. Sampling tests (see Table II, Groups 1 and 2).

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TABLE II. Quality conformance inspections - schedule and sequence[1/].

Tests	Test requirement paragraph	Test method paragraph	Sample			
GROUP 1			ALL SAMPLES			
Cleanliness	3.4.1	4.6.1	1			
Inspection	3.3.6	4.6.2	2			
Proof pressure	3.4.2	4.6.3.1	3			
Leakage	3.4.3	4.6.3.2	4			
Operating pressure	3.4.4	4.6.3.3	5			
Pressure drop	3.4.5	4.6.3.4	6			
GROUP 2[2/]			A	B	C	D
High temperature	3.4.6,1	4.6.4.1	1	x	1	x
Low temperature	3.4.6.2	4.6.4.2	2	x	2	x
Thermal shock	3.4.6.3	4.6.4.3	x	1	x	1
Dust	3.4.6.4	4.6.4.4	x	2	x	2
Salt fog	3.4.6.5	4.6.4.5	x	3	x	3
Humidity	3.4.6.6	4.6.4.6	x	4	x	4
Vibration	3.4.6.7	4.6.4.7	3	x	3	x
Service life[3/]	3.4.7	4.6.5	x	5	x	5
Burst pressure	3.4.8	4.6.6	4	x	4	x

[1/] Test sequence is denoted by the number in the sample columns. An X denotes test not required.
[2/] The procuring activity shall decide which tests of Table II, Group 2, each additional charging valve in excess of four shall receive see 6.2.1).
[2/] Charging valves subjected to life tests shall not be subjected to the burst test.

4.5.1 Individual tests. Individual tests are those tests conducted by the contractor on each charging valve submitted for acceptance under the contract or purchase order. Individual tests for charging valves are shown in Table II, Group 1. Failure to pass these tests shall be cause for rejection.

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4.5.2 Sampling tests. A random sampling of charging valves shall be selected by the procuring activity from each inspection lot in accordance with Table III and shall be subjected to the tests listed in Table II, Groups 1 and 2. Group 1 tests shall be conducted prior to Group 2 tests. Failure to pass these tests shall be cause for rejection of the lot. The procuring activity reserves the right to delete tests and alter the test sequence of Table II (see 6.2.1). In the event any sample representing an inspection lot fails to meet the inspection and test requirements outlined herein, the contractor, at the discretion of the procuring activity, may have the opportunity of reworking the entire affected lot (see 6.2.1). The procuring activity shall determine which inspections and tests of this specification are to be performed on the reworked inspection lot samples (see 6.2.1). The number of test samples shall be at the discretion of the procuring activity (see 6.2.1).

4.6 Inspection methods.

4.6.1 Cleanliness. The charging valve shall be subjected to the cleanliness test detailed herein. The test shall be conducted in a clean, dust-free atmosphere and all equipment shall be thoroughly cleaned. The charging valve shall be visually examined for cleanliness and shall be free of all water, oils, grease and foreign matter exceeding 0.002 inch in size on all surfaces internally interfaced with the nitrogen receiver,

4.6.2 Inspection. The charging valve shall be inspected as follows.

4.6.2.1 Individual tests-visual. The charging valve shall be inspected to ensure that it has been properly assembled, and that the workmanship is as specified in 3.6.

4.6.2.2 First article and quality conformance tests.

a. The charging valve shall be thoroughly examined to determine conformance to all the requirements of DL 58A164D556 and this specification.

TABLE III. Sampling test items.

Lot size	Number of samples to be tested
100 or less	2
101 to 300	3
301 to 500	4
501 or more	4 plus 1 for each additional 500 or fraction thereof.

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b. The charging valve shall be examined to ascertain that the packaging, packing and marking are in accordance with Section 5.

4.6.3 Functional tests. To perform the functional tests specified herein, requires the opening and closing of the charging valve. To open the valve, turn the hex nut counterclockwise. Take care not to jam the valve in the open position. To close the valve, turn the hex nut clockwise with a torque of 50 ±5 inch-pounds. During the closing procedure, a pressure of 75 ±25 psig shall be applied to the inlet or outlet port to ensure purging. The charging valve shall be subjected to the following tests.

4.6.3.1 Proof pressure. The charging valve shall be subjected to a proof pressure test. The outlet port of the charging valve shall be pressurized to 5500 ±100 psig under the following conditions:

a. Charging valve closed; inlet port opened to atmosphere.

b. Charging valve opened; inlet port sealed with the MS20813 cap supplied.

Each condition shall be maintained for three minutes. There shall be no visual indication of permanent deformation.

4.6.3.2 Leakage. The charging valve shall be checked for leakage with its outlet port pressurized to each of the following pressures for three minutes.

a. 200 psig

b. 1000 psig

c. 2200 psig

d. 3000 psig

e. 3800 psig.

Leakage shall not exceed 0.2 cubic centimeters per minute at any of the pressures between 200 and 3800 psig.

4.6.3.3 Operating pressure. The charging valve shall be subjected to an operating pressure test. The inlet port shall be pressurized to 3800 ±100 psig. The charging valve shall be slowly opened to allow pressurization of the fixture. The charging valve shall then be reclosed and the inlet pressure line removed. The valve shall operate properly and leakage shall not exceed 0.2 cubic centimeters per minute, with the 3800 psig remaining in the fixture.

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4.6.3.4 Pressure drop. The charging valve shall be subjected to a pressure drop test. With the charging valve fully opened, an inlet pressure of 3000 ±100 psig shall be maintained while flowing 28.3 liters per minute (1 cubic foot per minute). The pressure drop across the valve shall not exceed 10 psig.

4.6.4 Environmental tests. The charging valvq shall be subjected to the following environmental tests.

4.6.4.1 High temperature. The charging valve shall be subjected to the high temperature test in accordance with MIL-STD-810, Method 501, Procedure I, except the temperature shall be 121°C (250°F). While at temperature conditions, the charging valve shall be closed and the outlet port pressurized to and maintained at 3800 -200 Psig. At the completion of the test, the charging valve shall be visually Inspected for damage and tested in accordance with 4.6.3.2 and 4.6.3.3.

4.6.4.2 Low temperature. The charging valve shall be subjected to the low temperature test in accordance with MIL-STD-810, Method 502, Procedure I for a period of four (4) hours. The storage temperature shall be -54°C (-65°F). While at temperature conditions, the charging valve shall be closed, and the outlet port pressurized and maintained at 3800 ±100 psig. At the completion of the test, the charging valve shall be visually inspected for damage and tested in accordance with 4.6.3.2 and 4.6.3.3.

4.6.4.3 Thermal shock. The charging valve shall be subjected to the thermal shock test in accordance with MIL-STD-810, Method 503, Procedure I. Prior to being subjected to the temperature conditions, the charging valve shall be closed and the outlet port pressurized to 3000 ±100 psig. At the completion of the test, the charging valve shall be visually inspected for damage and tested in accordance with 4.6.3.2 and 4.6.3.3.

4.6.4.4 Dust. The charging valve shall be subjected to the dust test in accordance with MIL-STD-810, Method 510, Procedure I. While at test conditions, the charging valve shall be closed and the outlet port pressurized to 225 ±25 psig. The inlet port shall be capped, At the completion of testing, the charging valve shall be visually inspected for damage and tested in accordance with 4.6.3.2 and 4.6.3.3.

4.6.4.5 Salt fog. The charging valve shall be subjected to the salt fog test in accordance with MIL-STD-810, Method 509, Procedure I except the duration shall be 50 hours. While at test conditions, the charging valve shall be closed and the outlet port pressurized to 225 ±25 psig. The inlet port shall be capped. At the completion of testing, the charging valve shall be visually inspected for damage and corrosion, and tested in accordance with 4.6.3.2 and 4.6.3.3.

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4.6.4.6 Humidity. The charging valve shall be subjected to the humidity test in accordance with MIL-STD-810, Method 507, Procedure I, except four 24-hour cycles shall be substituted for the ten 24-hour cycles. While at test conditions, the charging valve shall be closed and the outlet port pressurized to 225 ±25 psig. The inlet port shall be capped. At the completion of testing, the charging valve shall be visually inspected for damage and corrosion, and tested in accordance with 4.6.3.2 and 4.6.3.3

4.6.4.7 Vibration. The charging valve shall be subjected to the following tests.

4.6.4.7.1 General. The vibration test specified herein shall be performed in each of three mutually perpendicular axes in the order shown. All testing shall be performed in one axis prior to testing in the next axis. As indicated herein, the charging valve shall be checked for proper operation.

4.6.4.7.2 Charging valve operation. While subjected to vibration, the charging valve shall be closed and the outlet port pressurized 3000 ±100 psig. The inlet port shall be capped. After each axis of vibration is completed, the charging valve shall be visually inspected for damage and tested in accordante with 4.6.3.2 and 4.6.3.3.

4.6.4.7.3 Mounting techniques. The chargin valve shall be attached to the vibration exciter table by means of the fixture described in 4.3.f. The vibration input shall be measured on the fixture at a point adjacent to the mounting point of the charging valve.

4.6.4.7.4 Sinusoidal test level. The charging valve shall be subjected to sinusoidal vibration along each of three mutually perpendicular axes. The frequency shall be ± percent and the amplitude shall be ±5 percent. The test shall consist of sweeping for 60 minutes in each, axis from t to 2000 Hz, at one octave per minute in accordance with Table IV.

TABLE IV. Sinsuoidal test level.

Frequency	Displacement or acceleration
5 - 20 Hz	0.100 inch da
20 - 33 Hz	±2.0 g's
33 - 74 Hz	0.036 inch da
74 - 2000 Hz	±10.0 g's

4.6.4.7.5 Random test level. The charging valve shall be subjected to a random vibration along each of three mutually perpendicular axes. The instantaneous random vibration acceleration peaks shall be limited to three times

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the rms acceleration level, The power spectral density of the test control signal shall not deviate from. the specified requirements by more than +100, -50 percent (±3 dB) between 20 Hz and 2,000 Hz, except that deviations as large as +300, -75 percent (±6 dB) shall be allowed over a cumulative bandwidth of 100 Hz, maximum, between 500 and 2,000 Hz.

Tolerance levels in terms of dB are defined as:

$$dB = 10 \log \frac{W_1}{W_0}$$

Where W_1 = measured acceleration power spectral density of g^2/HZ units.
The term W_0 defines the specified level in g2/Hz units.

Confirmation of these tolerances shall be made by use of an anlysis system providing statistical accuracies corresponding to a bandwidth-time constant product. BT = 50 minimum. Specific analyzer characteristics shall be as specified below or equivalent.

a. On-line, contiguous filter, equalization/analysis system having a bandwidth = B = 50 Hz maximum.

b. Swept frequency analysis systems characterized as follows:

(1.) Constant bandwidth analyzer.

(a) Filter bandwidth as follows:

B = 20 Hz, maximum between 20 and 200 Hz.

B = 50 Hz, maximum between 200 and 2,000 Hz.

(b) Analyzer averaging time = T = 2RC = 1 second, minimum, where T = true averaging time and RC = analyzer time constant.

(c) Analysis sweep rate [linear) = S = $\frac{B}{4RC}$ or $\frac{B^2}{8}$ Hz/ see, whichever is smaller.

(2.) Constant percentage bandwidth analyzer.

(a) Filter bandwidth= pf = one third octave minimum (0.23f_c) where p = percentage and f_c = analyzer center frequency.

(b) Analyzer averaging time = T = 50/pfc, minimum.

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(c) Analysis sweep rate (logarithmic) = S = $\frac{pf_c}{4RC}$, or $\frac{(pfc)2}{8}$ Hz/second, whichever is smaller.

c. Digital-power, spectral-density analysis systems employing quantization techniques shall. provide accuracies corresponding to the above approach.

The random vibration test level shall be a flat spectrum in the range of 20 to 2000 Hz at a PSD level of 0.02 g^2/Hz with an overall test level of 6.3 g's rms. The test time shall be 40 minutes in each of the three mutually perpendicular axes. The analysis shall be as shown in Table V.

TABLE V. Vibration analysis.

Bandwidth	Analysis filter
20 - 200 Hz	20 Hz filter
200 - 2000 Hz	50 Hz filter

The analysis shall yield a statistical accuracy of 120 degrees of freedom.

4.6.5 Service life. The charging valve shall be subjected to a service life test of 500 operating cycles, each cycle consisting of an opening and closing sequence as described herein and in accordance with the precautionary note of 4.6.3. The inlet port shall be pressurized to 3000 ±100 psig. The charging valve shall be slowly opened to allow pressurization of the fixture. The charging valve shall then be reclosed and the inlet pressure line vented to atmosphere. The fixture shall then be vented to a pressure below 2000 psig. At the completion of each 50 cycles, the charging valve shall be visually inspected for damage and tested in accordance with 4.6.3.2 and 4.6.3.3.

4.6.6 Burst pressure. The charging valve shall be subjected to a burst pressure test. The charging valve shall be closed, 'but the inlet port shall remain uncapped. The outlet port shall be hydrostatically pressurized to 6700 ±100 psig for a period of three minutes. There shall be no evidence of permanent deformation.

5. PREPARATION FOR DELIVERY

5.1 Charging valve preservation and packaging.

5.5.1 Level C. Each charging valve inlet port shall be protected with a cap free from all contaminants and conforming to MS 20813. Charging valves

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shall be individually packaged in a heat-sealable bag conforming to MIL-B-22020, Type I. The charging valve shall be packaged free of oil, water and water vapor and the bag shall be dust-free, lint-free and free from water vapor when sealed.

5.2 Packing.

5.2.1 Level C. Charging valves shall be packed in accordance with MIL-STD-794.

5.3 Marking. All unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES.

6.1 Intended use. The charging valve is a positive closure for a nitrogen receiver which mechanically opens to permit gas flow for filling or venting the nitrogen receiver.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number, revision letter and date of this specification.

b. Government-approved test laboratory (see 3.1.2).

c. First article requirement (see 3.1 and 3.1.2).

d. Option for removal from AS/QVPL (see 3.2.4.3).

e. Option for quality reverification (See 3.2.4.3).

f. Responsibility for inspection (see 4.1). -

g. Samples subjected to sampling inspection (see 4.5.2) shall not be considered or accepted as part of the contract or purchase order.

h. Rework and retest requirements and quantity to be submitted for inspection lot samples (see 4.5.2).

i. Tests deleted and altered test sequence (see 4.4, 4.5.2 and footnote 2/ on Table III.

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and involves the provisions of 7-104.9(n) of the Defense Acquisition Regulations, the data requirements identified below

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will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in "the following paragraphs:

Paragraph	Data requirement	Applicable DID
3.2	Test Reports	DI-T-21466A
3.2.1, 3.2.2 3.2.4 & 3.2.4.2	Approved Parts Source	UDI-R-20331B
3.2.4.1	Test Procedures	DI-T-3702A
3.3	Engineering Change Proposal	DI-E-2037
4.4.1	Report, First Article Test	UDI-T-23790

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Quality assurance definitions. Definitions for quality assurance terms are in accordance with MIL-STD-109.

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The first article should be a first production item. The first article should consist of four charging valves. The contracting officer should include specific instructions in all procurement instruments, regarding arrangement for examinations, test and approval of the first article.

6.5 Contract or purchase order provision. Contract or purchase orders should specify the following provision for first article inspection. "When a contractor is in continuous-production of the charging valve applicable to this specification from contract to contract, consideration shall be given to waive the first article inspections. However, waiver of first article inspections shall not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract." The words "purchase order" should be substituted throughout for the word "contract" when applicable.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy-AS

(Project 4820-N428)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-V-85033A(AS) VALVE, CHARGING

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

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